

STÉPHANE BANCELIN

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CNRS RESEARCHER IN BIOPHOTONICS

SCIENTIFIC AREAS

- Light/Matter interaction, biophotonics, nonlinear optics, neuroscience
- Optical microscopy: multiphoton (SHG, CARS, THG, SFG/DFG), super-resolution (STED), polarimetry, ellipsometry, interferometry, adaptive optics
- Computational analysis: 3D reconstruction, segmentation, morphological mathematics, numerical modelling
- Collagen chemistry: protein aggregation (fibrillogenesis), extraction/purification, hydroxyproline dosing
- Tissue manipulation & imaging: skin, brain, bone, cartilage, cornea, embryos, tendon, biomimetic materials
- Tissue physiopathology: skin biomechanics, prolapses, cancer, embryogenesis, EDS, osteoarthritis, Alzheimer
- Animal experiments: in vivo imaging, brain surgery, stereotaxic injection, animal care

RESEARCH EXPERIENCES

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| <i>Since 2023</i> | CNRS Researcher
Laboratoire de Photonique Numérique et Nanosciences (LP2N), CNRS - Institut d'Optique - Univ. de Bordeaux
<i>Non-linear microscopy to study connective tissue biomechanics.</i> |
| <i>2020-2022</i> | CNRS Researcher
Institut Interdisciplinaire de Neurosciences (IINS), Université de Bordeaux - CNRS
<i>Adaptive optics for in vivo STED microscopy.</i> |
| <i>2016 - 2019</i> | Post-doctoral researcher with Prof. V. Nägerl
Institut Interdisciplinaire de Neurosciences (IINS), Université de Bordeaux - CNRS
<i>In vivo super-resolution STED imaging of dendritic spines in the hippocampus.</i> |
| <i>2014 - 2016</i> | Post-doctoral researcher with Prof. F. Légaré
Institut National de la Recherche Scientifique, Centre Energie Matériaux et Télécommunication, Université du Québec
<i>Interferometric Second Harmonic Generation microscopy for non-invasive tissue imaging.</i> |
| <i>2010 - 2013</i> | PhD in Biophotonics under the supervision of Dr Marie-Claire Schyanne-Klein
Laboratoire d'Optique et Biosciences (LOB), École Polytechnique - CNRS – Inserm |

Quantitative imaging of collagen using Second Harmonic Generation microscopy.

- 2010 **Master internship** with Dr K. Plamann
Laboratory for Applied Optics (LOA), ENSTA ParisTech - CNRS
Study of corneal transparency
- 2009 **Master internship** with Dr David Neely
Rutherford Appleton Laboratory (RAL), University of Oxford
Protection of the optics in the HIPER inertial fusion reactor
- 2008 Bachelor internship with Anraud Landragin
Laboratoire de référence Temps/Espace (SYRTE), Observatoire de Paris
Implementation of an extended cavity laser : application to cold atoms manipulations

TRAINING

- 2013 **PhD in Biophotonics**
Ecole Polytechnique
- 2013 **University degree of advanced biomedical studies**
IFSBM - Paris Sud University
- 2010 **MSc in Optics** with a specialty in Laser/Matter interactions
ENS Cachan - Ecole Polytechniques - Paris VI University (UPMC)
- 2008 **BSc in experimental physics**
ENS Cachan - Paris VI University (UPMC)

TEACHING

- Since 2020 **Coordinator of the IINS-BIC Labcourse** for the M2 EUR-Light program at the University of Bordeaux (2 weeks/year)
- 2021-2024 **Master's courses** : EUR Light et EUR Neurophotonic (6 h/year).
Super-resolution STED microscopy in neurosciences et Microscopy beyond fluorescence.
- Since 2023 **Bibliographic projects** at the Institut d'Optique (1 week/year)
- 2018 - 2023 **Instructor for the National Training Action** - Bordeaux Imaging Center (~4 h/year)
Microscopie de super-résolution.
- 2017 - 2022 **Instructor at the CAJAL School**, Bordeaux School of Neuroscience (2 weeks/year).
Connectomics : from micro to meso and macro-scale, Linking neuronal circuit and behavior, Advanced imaging techniques for cellular and systems neuroscience.
- 2010 - 2013 **Teaching assistant** in physics, Paris-Sud Faculty of Medecin (64h/year)
- 2008 - 2010 **Oral examiner** for the preparatory class for Engineer school, PSI* track - Lycée Jean-Baptiste Say (Paris) (3h/week)

STUDENT'S SUPERVISION

- **PhD students**
- Since 2026 **Florestan Luga**, PhD student at LP2N, Co-direction with F. Légaré at the University of Quebec
Correlative SHG - Brillouin microscopy in articular cartilage.
- Since 2025 **Théo Martinaud**, PhD student at LP2N
Nonlinear optical microscopy to visualize the architecture of connective tissues.
- Since 2023 **Nasim Akhtar**, PhD student at LP2N, Co-direction with L. Cognet
Multimodal 3-photon fluorescence microscopy applied to neurophotonics.
- Since 2023 **Rohit Mangalwedhekar**, PhD student at LP2N, Co-direction with L. Cognet
Super-resolution multimodal dynamic microscopy in biophysical models and biological tissues.

- 2020 - 2023 **Johannes Roos**, PhD student at IINS - Co-supervision with V. Nägerl
Arkitekt: An open-source framework for modern bioimaging workflows.
⇒ Currently post-doctoral researcher at Johannes Kepler University of Linz
- **Undergrad student**
- 2025 **Théo Martinaud**, Master's student at LP2N
Implementation of a VIPA spectrometer and integration into a scanning microscopy.
- 2025 **Matys Lambert**, 3rd- year in the IUT Physical measurements program
Building of a symmetric tensile extensometer to study connective tissues.
- 2023 **Gioele Sassi**, M1 student at LP2N - currently PhD student at the University of Trento
Design and implementation of a microscope for Multiphoton imaging of Single-Walled Carbon Nanotubes in IR range.
- 2022 **Mohamed Belkadi**, M2 student at IINS - currently engineer at Belimo
PSF optimization using SLM-based adaptive optics in brain tissue.
- 2021 **Philippe Nguyen**, M2 student at IINS - currently PhD at CRPP
Development of quantitative approaches for inverse-contrast STED microscopy.
- 2019 **Sabin Sapkota**, Bachelor student at IINS
Measuring laser pulse duration in a STED light using Michelson interferometer.
- 2018 **Carlotta Pessina**, M2 student at IINS - currently lead consultant at BIP
2P-STED microscopy and aberration correction through adaptive optics for neuroscience applications.
- 2017 **Rached Baida**, M1 student at IINS - currently engineer at Seemens Healthinners
Confocal microscopy with a Tip-tilt scanner.
- 2015 **Etienne Auger**, Bachelor student at INRS-EMT - currently engineer at IR&T
Interferometric SHG imaging of equine cartilage.
- 2014 **Katherine Légaré**, Bachelor student at INRS-EMT - currently researchers at TeraXion
Image analysis in interferometric SHG microscopy.

COLLECTIVE RESPONSIBILITIES & REVIEWING ACTIVITIES

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- 2026 Expert evaluator for Horizon Europe – Doctoral Network
- Since 2025 LP2N Lab council member
- 2024 - 2025 Organizer of the LP2N microscopy Journal club
- Since 2024 Member of the AI comitte, deployment of a local AI on the lab's computing server
- 2024 Project evaluator - Envergure-Amorçage calls for proposals: Région Bourgogne Franche Comté
- 2024 Project evaluator - ANR PRC
- 2024 Organizer of the LP2N laboratory retreat (~80 participants)
- 2023 - 2024 Coordinator of LP2N student seminars
- 2023 Reviewer for the Pierre Favard prize – French society of microscopy
- 2023 Reviewer for the « Espoir en tête » exceptional project call - FRC
- 2022 Reviewer for the AAP Research Equipement program - A. Midex
- Since 2021 Expert evaluator for Horizon Europe - Marie Skłodowska-Curie Action (MSCA)
- 2019 - 2022 Module coordinator at the CHRONIC animal facility at Neurocampus and member of the SBEA animal welfare committee for the PIV-EXPE and CHRONIC animal facilities
- Since 2015 Reviewer for international peer-review journals
Nature Communication, Optica, Scientific Reports, Biomedical Optics Express, Optics Letters, Optics Express, International Journal of Biological Molecules, Revue Française d'Histotechnologie, NanoLetters, and Scientific Data.
- 2011 - 2013 Elected member of the LOB laboratory council
- 2010 Organizer of the 22nd IFSBM Symposium
Nanotechnologies and Biomedical applications (150 participants)

FUNDINGS & DISTINCTION

2026	SMR Departement Call for proposals - 1 year - 7.5k€ Role: PI <i>BriTis: Brillouin microscopy to characterize soft connective tissues biomechanics.</i>
2024	New Aquitaine Research Call for proposal - 3 years - 192 k€ Role : PI Partner : Poietis <i>MOMI : Multimodal optical microscopy to probe the architecture of bioprinted tissues.</i>
2023	ANR JCJC - 5 ans - 349 k€ Role : PI <i>HaBlm: Second Harmonic and Brillouin imaging to probe multiscale vocal cords biomechanics.</i>
2023	Emergence INP - 1 year - 60 k€ Role: PI <i>Setup of a 1700 nm multiphoton microscope for imaging biological tissues.</i>
2023	SMR Departement Call for proposals - 1 year - 10k€ Role: PI <i>BioMulR: Assembly of a microscope-mounted extensometer for studying the biomechanics of connective tissues.</i>
2023	GPR Light - PhD grant for Rohit Mangalwedhekar - 3 years - 115 k€ Role: co-PI with Laurent Cognet (50%) <i>RoMan: Multimodal dynamic super-resolution microscopy in biophysical models and biological tissues.</i>
2020	ERC Synergy - 6 years - 10 M€ PI: Laurent GROC Partners: Lauren COGNET, Erwan BEZARD, Valentin NÄGERL Role: Task leader in Valentin Nägerl during 2021-2022 and in Laurent Cognet since 2023. <i>ENSEMBLE: Structure and function of the brain extracellular space</i>
2018	Marie Curie Individual Fellowship - 2 years - 185 k€ <i>IVSTED: in vivo super-resolution imaging of synapses in the hippocampus.</i>
2017	AXA Research Fund postdoctoral grant - 2 years - 130 k€ <i>Super-resolution STED microscopy in the hippocampus in vivo to reveal changes in synapses underlying memory impairment in Alzheimer's disease.</i>
2014	RTRA Physics Triangle best PhD award
2010	MERNT doctoral grant - 3 years - 90 k€ <i>Quantitative imaging of collagen using Second Harmonic Generation microscopy.</i>